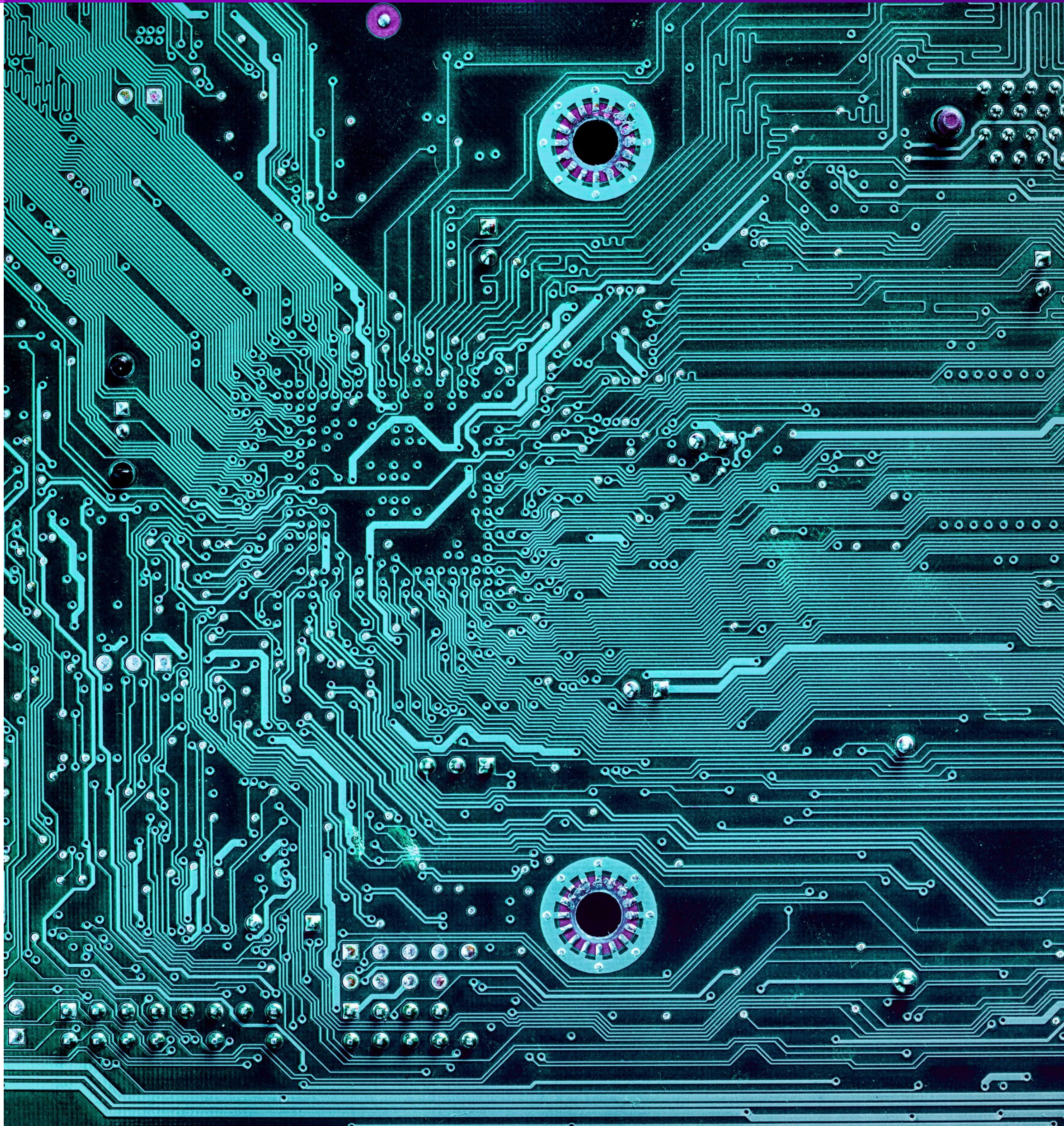
Course Introduction

MTE 220 – Sensors and Instrumentation

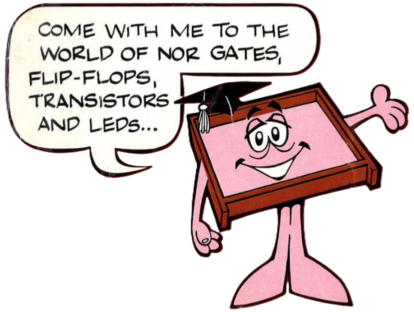


UNIVERSITY OF WATERLOO | FACULTY OF ENGINEERING

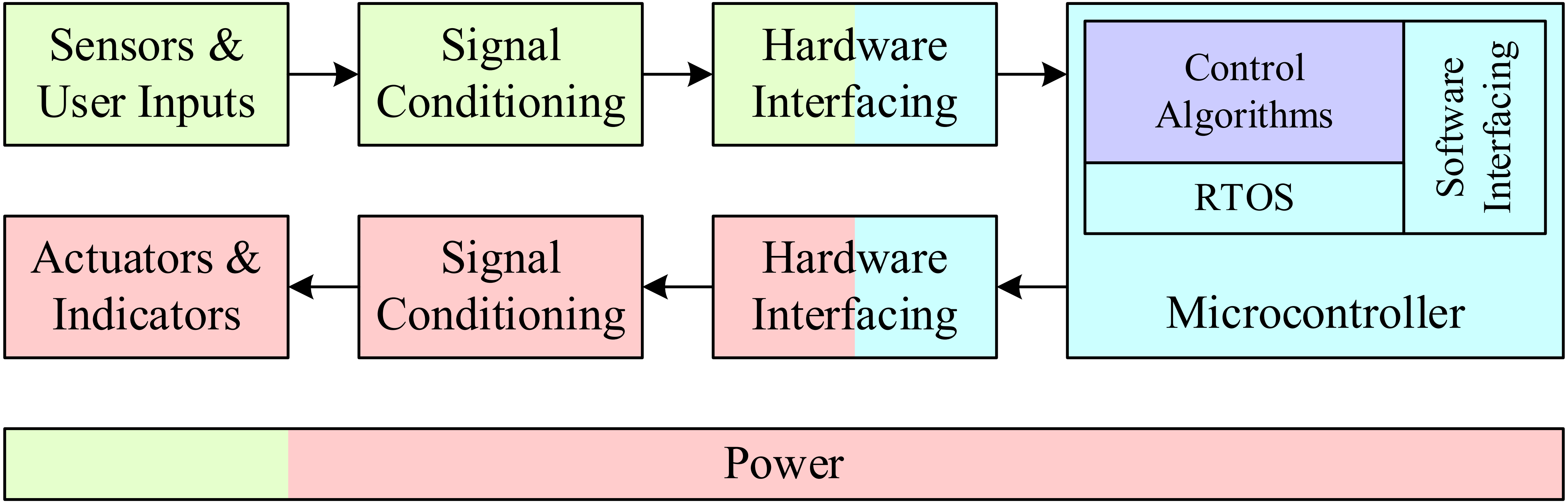
Welcome!

Course Instructors

- I'm an electrical and biomedical engineer
- I love circuits!
- I've been at UW in ECE since 2015
- I do a *lot* of administration because of my business background
- My research is in neuromorphic circuits and also multidomain modeling and simulation



MTE's Automation and Control Pathway



- MTE 220 (2B): Sensors and Instrumentation
- MTE 320 (3A): Actuators and Power Electronics
  MTE 420 (4A): Power Electronics and Motor Drives
- MTE 360 (3B): Automatic Control Systems
  ECE 484 (4A): Digital Control Applications
- MTE 262 (2A): Microprocessors and Digital Logic
  MTE 325 (3A): Microprocessor Systems and Interfacing
  MTE 241 (2B): Introduction to Computer Structures & Real-Time Systems

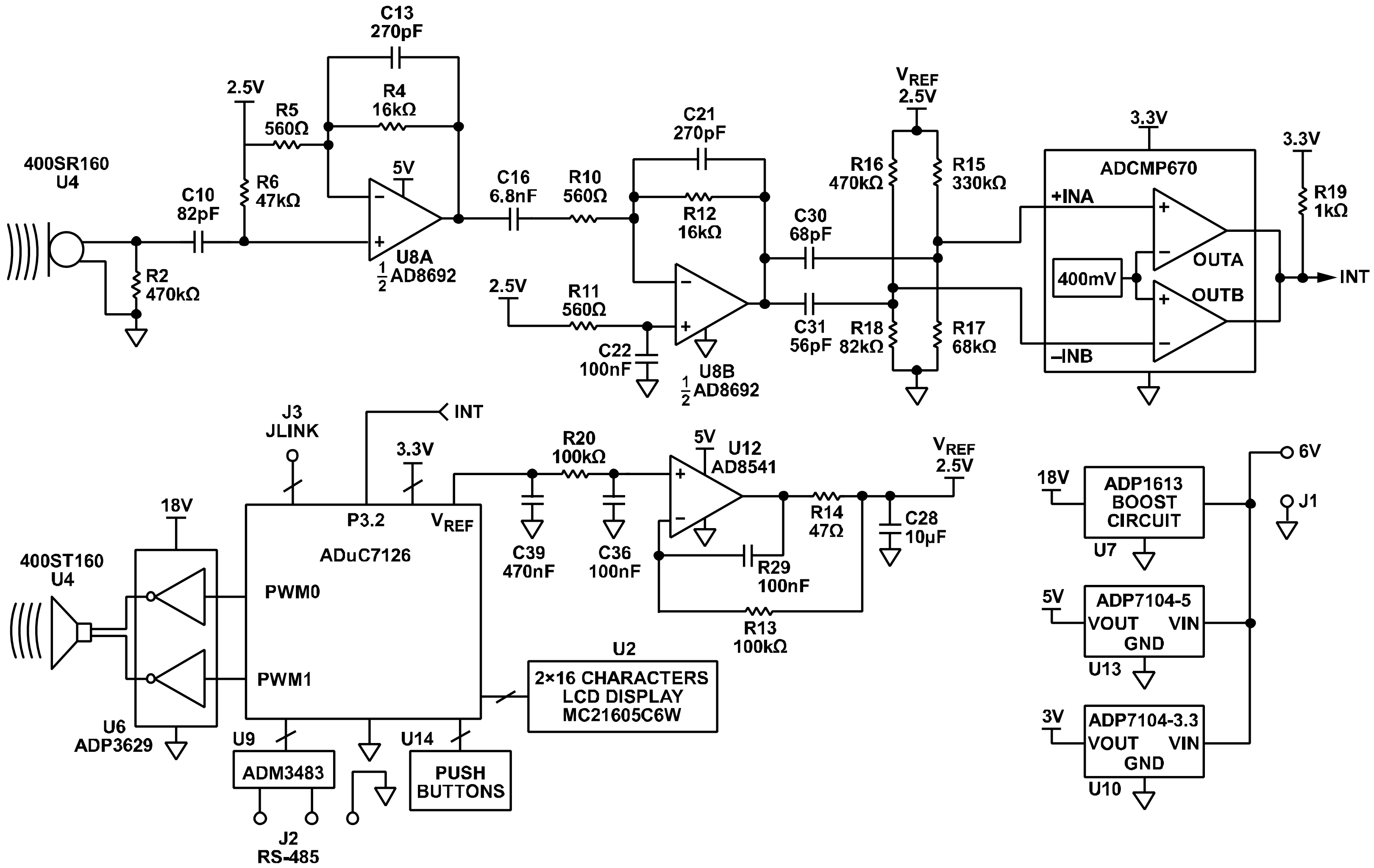
Course Outcomes

By the end of this course, you should be able to:

- **Analyze** circuits consisting of linear and nonlinear components and sensors
- **Design** signal conditioning circuits based on given requirements
- **Understand** basic embedded system power supply options
- **Use** appropriate lab equipment to **build**, **debug**, and **characterize** circuits for mechatronics systems

Practical Example of Course Outcome

By the end of this course, something like this will make sense:



Grading Scheme

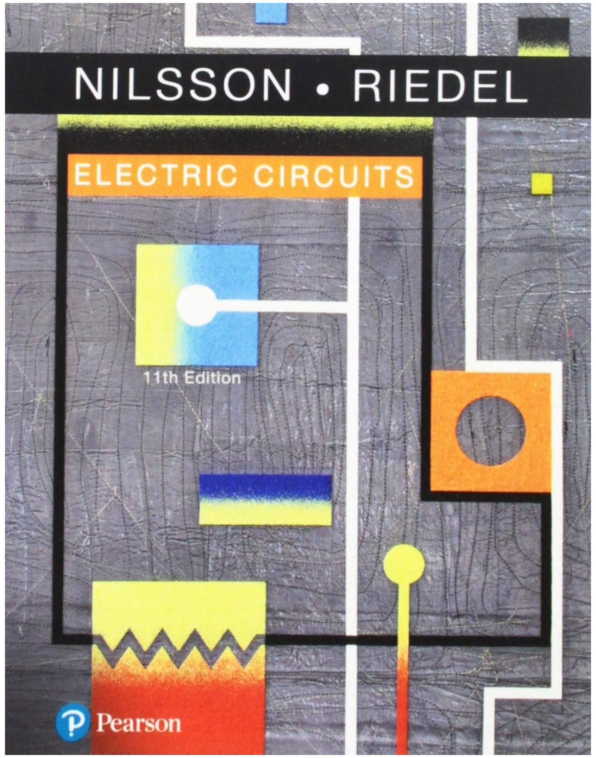
Item	Weight	Notes
Labs	25%	Labs 1 – 4
Project	10%	
Midterm Exam	25%	The Final Exam is worth 65% if you miss the Midterm Exam
Final Exam	40%	All midterm exam weight shifted to the Final Exam if you show improvement

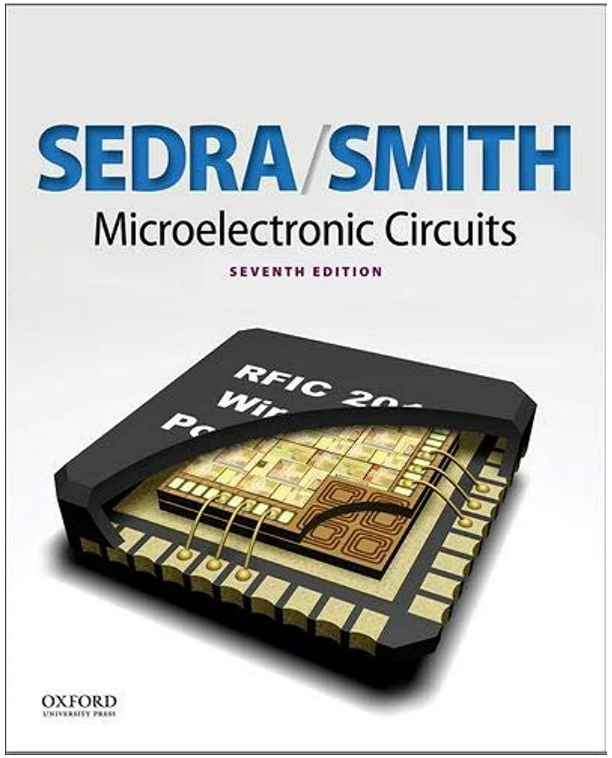
Course Resources

- LEARN
  - I will post announcements here
  - Course notes, blank and annotated lectures, lab information, quizzes, etc.
- MS Teams
  - The Team's calendar includes all lectures, tutorials, and exams
  - Recorded lectures and tutorials available as recorded meetings
- Crowdmark
  - The midterm and final will be in person and graded on Crowdmark
- Piazza

Textbooks

- Optional
  - I believe the Course Notes, Lecture Slides, and provided textbook excerpts are sufficient, but…





Topics

1 – Signals and Interfaces

2 – New Devices and Circuits

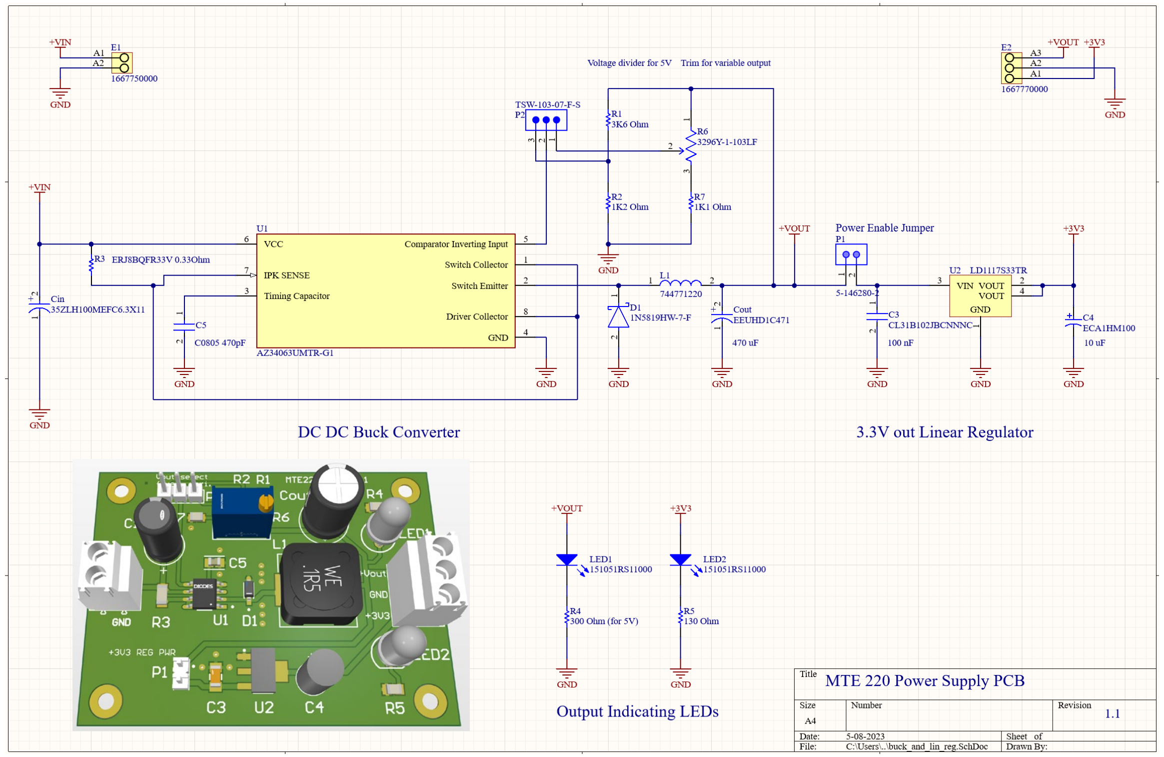
3 – Sensors

4 – Interfacing and Design

Labs

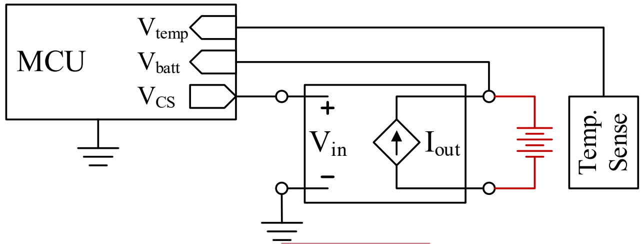
- Learn SMT and Through-Hole PCB assembly
- Learn how to prototype on breadboards and PCBs
- Learn how to calibrate and use sensors
- Learn how to power embedded systems

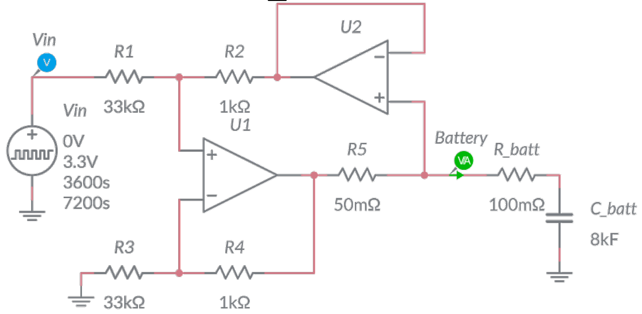
Lab	Activity
1	Current Sensing
2	Hall Effect Sensor & Schmitt Trigger
3	Power PCB
4	PWM + Filter
5	Project Demos

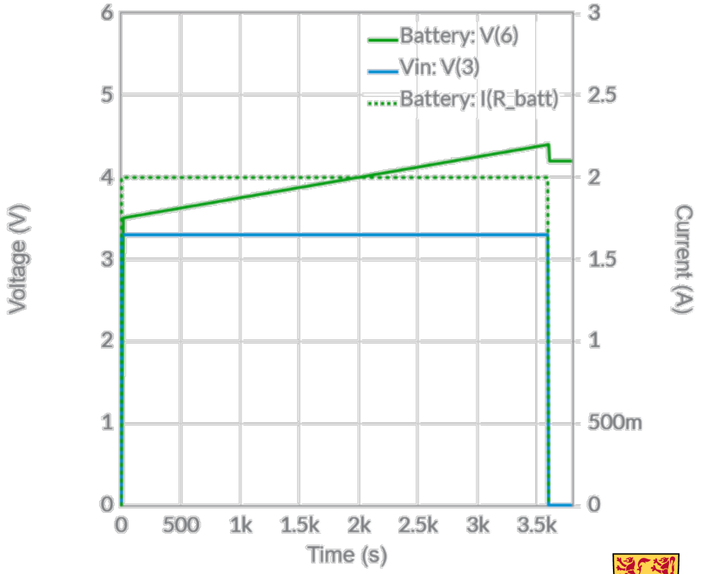


Project

- Explore a circuit beyond those covered in lectures or labs
- Connect theory to real-world applications
- Strengthen your ability to explain engineering work clearly and confidently
- Demonstrate independent learning and curiosity in engineering practice
- Learn LTSpice well enough to put on your resume
- Project scales to interest level:
  Demonstrate learning regardless of starting point

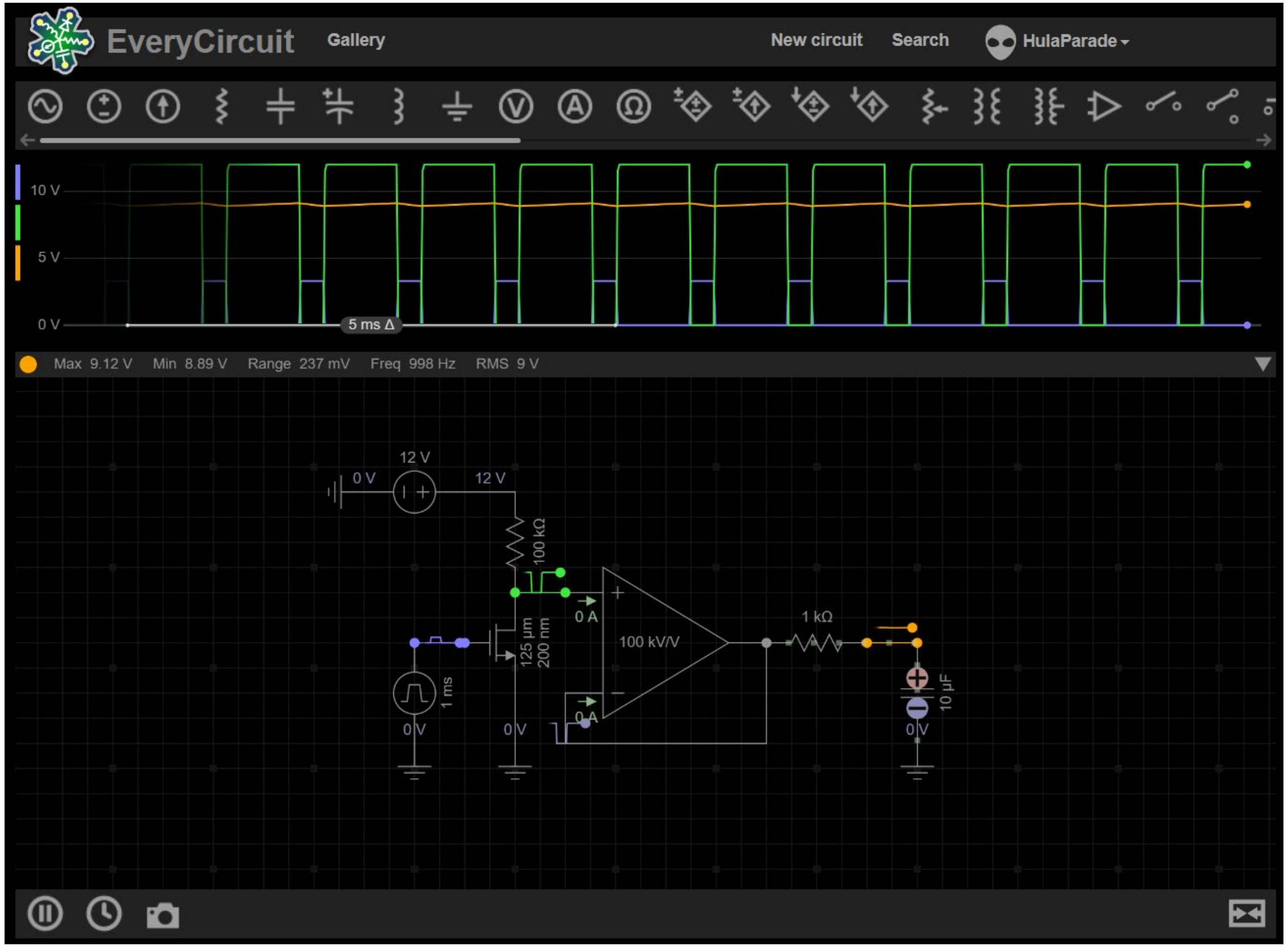






Simulator: EveryCircuit

- https://everycircuit.com/app
- Search @HulaParade
- Used in-class to illustrate various concepts



Simulator: LTSpice

- https://www.analog.com/en/resources/design-tools-and-calculators/ltspice-simulator.html
- Used for the project and more complex circuits
- Everyone will need to use this